

# EASI-Mem: Evidence-Aligned Safe Policy Evolution for Budgeted Long-Dialogue Memory

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## Abstract

Long-horizon agents need memory policies that decide how many candidates to retrieve and how much memory to inject, yet end-to-end task accuracy entangles memory with the reader model, tools, prompting, and decoding. We present EASI-MEM, an evidence-aligned, bounded policy-evolution sidepath for the open-source MemoryCore system. It preserves the existing FTS/vector substrate, scores injected provenance units directly against public evidence annotations, optimizes only a finite set of candidate and token-budget profiles, and promotes a policy only when paired confidence and cost guards pass. Missing, corrupt, incompatible, timed-out, or failed policies execute the exact base path.

On a 98-case LongMemEval-S held-out split, the development-selected policy changes macro evidence-session recall from 82.40% to 85.00% while reducing mean injected tokens by 30.09%. The recall-delta 95% interval is  $[-0.51, +6.07]$  percentage points, satisfying a predeclared 1-point non-inferiority guard, and the joint utility interval is positive. We do not present this as pristine confirmation because earlier protocol iterations had exposed aggregate LongMemEval test results. More importantly, a checksum-frozen transfer to 1,536 evidence questions in LoCoMo fails: tokens fall 27.56%, but macro session recall falls 4.52 points with a conversation-cluster interval of  $[-5.61, -3.35]$ . An exploratory leave-one-conversation-out calibration rejects every candidate and selects the base path in 10/10 folds, preventing the harmful update but producing no gain. These results establish a useful boundary: direct evidence feedback can support cheap, auditable optimization and safe rejection, but a budget policy learned on one dialogue distribution is not universally transferable. The release includes dataset adapters, versioned protocols, structured receipts, provenance persistence, a production switch, and forced-failure tests suitable for later connection to internal multi-turn programming data.

**Keywords:** agent memory, long dialogue, self-optimization, retrieval evaluation, provenance, safe fallback, token budget

## 1 Introduction

Persistent agents increasingly separate a bounded prompt from a larger memory substrate. MemGPT frames this as virtual context management [8]; systems such as Zep and Hindsight add temporal graphs, typed memories, and reflection [5, 9]. Open-source MemoryCore provides a layered substrate (raw L0 conversation, L1 atomic memory, L2 scenes, and L3 profiles), SQLite FTS5, vector search, fusion, extraction, and injection [11]. In such systems, however, *policy* is commonly static: Top- $k$ , candidate depth, and injection budget are fixed at deployment.

Self-optimizing memory work makes this policy adaptive. Memory-R1 learns memory operations through reinforcement learning [13]; EvolveMem searches an explicit configuration space with regression rollback [6]; and SelfMem lets an agent refine its memory strategy from environmental feedback [14]. These approaches motivate policy evolution, but they leave a practical systems question: how can an existing memory service evolve a small, auditable policy without retraining or replacing its retrieval core, and how can it refuse an update when the feedback distribution shifts?

End-to-end answer accuracy is an expensive and noisy optimization target. A failure may come from retrieval, reasoning, a tool, prompt construction, or decoder variance. Conversely, a correct answer does not prove that injected memory helped. LongMemEval and LoCoMo provide evidence annotations that permit a more direct necessary-condition test: whether the memory boundary supplied the labeled source units [7, 12].

We make four contributions:

1. We define a provenance-aligned evaluation protocol that scores evidence sessions and turns separately from answer generation, together with exact injected-token and retrieval-latency costs.
2. We introduce a bounded evolution layer over the unchanged MemoryCore retriever. A finite profile space, shallow confidence-gated router, versioned policy schema, and statistical promotion gate make adaptation reviewable.

3. We implement a disabled-equivalent production sidepath with hard fallback and structured decision logs, plus L1 provenance persistence across SQLite and Tencent Cloud VectorDB paths.
4. We report both positive developmental and negative external evidence. The latter reveals a granularity-induced transfer failure and demonstrates why safe rejection is part of optimization rather than an implementation afterthought.

Our scope is deliberately narrow. Public long-dialogue data validates the method; it does not establish performance on real long programming tasks.

## 2 Related Work

**Agent memory systems.** The field spans token-level, parametric, and latent forms, with factual, experiential, and working-memory functions [3]. MemGPT manages tiers around a finite context window [8]. Zep’s Graphiti component maintains a temporal knowledge graph [9], while Hindsight separates facts, experiences, observations, and beliefs and exposes retain/recall/reflect operations [5]. These systems improve memory representation and lifecycle behavior. EASI-MEM instead holds representation and the base retriever fixed so that policy effects remain attributable.

**Learned and evolving policies.** Memory-R1 learns ADD, UPDATE, DELETE, and NOOP decisions and answer-side memory selection [13]. EvolveMem uses failure logs and a meta-analyzer to modify retrieval configuration with revert-on-regression [6]. SelfMem exposes tools and feedback that permit broad strategy refinement [14]. Our approach is less expressive: it searches a pre-reviewed finite space and may emit only a bounded tree or global leaf. The reduction in expressivity is intentional: it permits static validation, exact fallback, and paired evaluation without model training.

**Long-term memory evaluation.** LongMemEval tests extraction, multi-session reasoning, temporal reasoning, updates, and abstention over scalable chat histories [12]. LoCoMo supplies long multi-session conversations and dialogue-level evidence annotations [7]. Their standard QA evaluations measure useful end behavior, whereas our primary metric stops at the injection boundary. This removes reader and judge variance but measures a necessary, not sufficient, condition for correctness.

**Retrieval infrastructure.** MemoryCore’s local substrate uses SQLite FTS5/BM25 [10]; its broader hybrid path can combine rankings using reciprocal rank fusion [1]. Our public experiment invokes the actual MemoryCore L0 FTS5 path. We do not claim to improve BM25, FTS5, embeddings, or RRF.

## 3 Problem Formulation

For query  $q_i$ , let  $G_i^S$  and  $G_i^T$  denote gold evidence sessions and turns. A base retriever returns ordered candidates  $C_i^{(K)}$  up to candidate limit  $K$ . A policy  $\theta$  selects a result limit  $k$ , token budget  $B$ , and bounded reranking weights, producing injected units  $I_i(\theta)$ .

Our primary effectiveness statistic is macro evidence-session recall:

$$R_S(\theta) = \frac{1}{N} \sum_{i=1}^N \frac{|S(I_i(\theta)) \cap G_i^S|}{|G_i^S|}, \quad (1)$$

where  $S(I)$  extracts provenance session identifiers. We analogously compute turn recall, any/all recall, and session NDCG [4]. Cost is the exact `c1100k_base` token count  $T_i(\theta)$  plus observed backend query latency.

For comparison policy  $a$  and base  $b$ , per-case utility is

$$u_i(a, b) = R_{S,i}(a) - R_{S,i}(b) + \lambda \frac{T_i(b) - T_i(a)}{\overline{T}(b)}, \quad \lambda = 0.03. \quad (2)$$

The small cost coefficient cannot compensate for a large recall loss; recall also has independent guards. A candidate is promoted only if:

$$\text{LCB}_{.95}(\Delta R_S) \geq -0.01, \quad (3)$$

$$\Delta R_{\text{any}}, \Delta R_{\text{all}} \geq -0.01, \quad (4)$$

$$1 - \overline{T(a)}/\overline{T(b)} \geq 0.10, \quad (5)$$

$$\text{UCB}_{.95}(\Delta T) < 0, \quad (6)$$

$$\text{LCB}_{.95}(\Delta u) > 0. \quad (7)$$

The evolution boundary is candidate/injection policy only. Storage semantics, FTS5/vector kernels, and the extraction model are outside the action space.

## 4 Method

### 4.1 Evidence-Aligned Adapter

Every dataset adapter emits the same case object: stable case and dependency group IDs; a query; ordered user/assistant turns with stable turn and session IDs; gold turn/session sets; category; and a negative-load flag. The runtime policy cannot receive answer, category, split, or gold labels.

LongMemEval’s `has_answer` turns define  $G_i^T$ , and `answer_session_ids` define  $G_i^S$ . LoCoMo dialogue IDs in `evidence` are mapped to their containing sessions. We append the dataset-provided image caption and image-search query to image-bearing LoCoMo turns, because some annotated facts are otherwise absent from text. Category-5 adversarial questions and four questions without evidence are negative-load probes, not positive recall cases.

### 4.2 Real MemoryCore Backend

For every case, all turns are inserted through MemoryCore’s SQLite-backed L0 schema. The runner calls the real `executeConversationSearch` function and FTS5 index. Candidate IDs are original turn IDs; session IDs and timestamps are preserved. This avoids a mock retriever that could accidentally encode labels.

We use L0 to isolate retrieval and injection. Evaluating L1 simultaneously would add extraction-model and asynchronous-pipeline variance. To make later L1 evaluation possible, the implementation persists source-message provenance through SQLite migrations/upserts/reads/FTS hydration, Tencent Cloud VectorDB, and memory-search responses.

### 4.3 Bounded Profile Space

The v1.4 space contains eight profiles. BASE returns Top-5 with no adaptive token cap. `fixed-896` requests 30 candidates, returns at most 10, and caps injection at 896 tokens. Six relative profiles use candidate limit 30, result limit 10, and ratio  $r \in \{.70, .75, .80, .85, .90, .95\}$ :

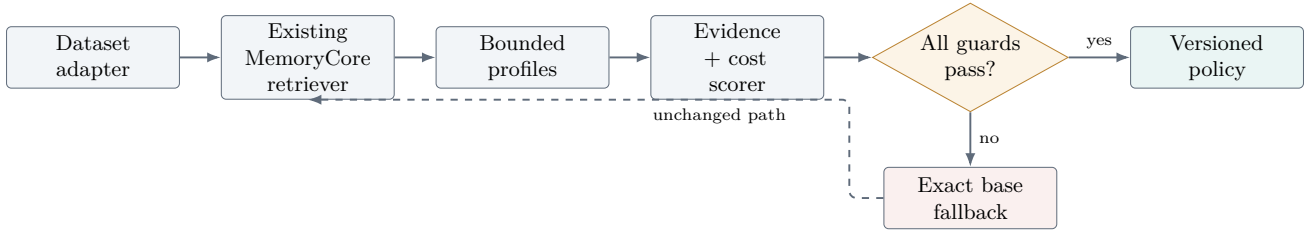
$$B_i(r) = \min(1024, \lfloor rT_i(\text{BASE}) \rfloor). \quad (8)$$

A greedy packer skips candidates that do not fit. Before packing, lexical near-duplicate similarity may reduce a candidate’s score by at most 12%; a recency multiplier is structurally capped at 10%. The promoted v1.4 profiles use no recency modulation.

The policy validator allows at most eight profiles, candidate limit 50, result limit 12, budget 4,096, ratio  $[.1, 1]$ , diversity weight .25, recency weight .10, and router depth five.

### 4.4 Evolution and Promotion

Training cases receive a label from the best profile under a lexicographic quality order (all-session recall, macro recall, any-session recall, NDCG, then tokens). A shallow decision tree can split on query length/cues, corpus size, and bounded Top-5 score, source-diversity, and token statistics. A leaf whose majority confidence is below a configured threshold selects the globally safe fallback. Global-profile and contextual-router promotion are evaluated separately.



**Figure 1:** The evolution loop changes only reviewed policy objects. The base retriever remains available and is the terminal path for rejection or runtime failure.

**Table 1:** Public data receipts.

| Dataset               | Immutable identifier                              | Scored | Groups           |
|-----------------------|---------------------------------------------------|--------|------------------|
| LongMemEval-S-cleaned | HF rev. 98d7416c...d48f; SHA-256 d6f21ea9...3a442 | 470    | 470 cases        |
| LoCoMo-10             | official file; SHA-256 79fa87e9...98ff4           | 1,536  | 10 conversations |

Across v1.2–v1.4, contextual routers were rejected; v1.4 therefore emits a static `RELATIVE-75` leaf. The controller detects that no scout features are needed and performs only one Top-30 query.

## 4.5 Production Execution and Fallback

The production option is nested under `recall.adaptive` and defaults to off. When off, auto-recall invokes the pre-existing search function directly. When on, a read-only loader validates the versioned policy and verifies that the runtime base profile matches the policy’s evaluated baseline. The existing retriever supplies candidates; the sidepath reranks and packs them; existing character budgets remain an additional guard.

The decision log records mode, policy revision, profile, candidate and result limits, effective token budget, returned items/tokens, fallback flag/reason, and latency. A missing or malformed file, invalid schema/limit, baseline mismatch, timeout, scout failure, or candidate-stage failure calls the exact base query. The implementation does not delete or permanently replace an index. A timed-out underlying call cannot be canceled if that retriever exposes no abort interface, but its result is never injected.

# 5 Experimental Protocol

## 5.1 Data and Splits

Table 1 gives immutable identifiers. LongMemEval-S-cleaned has 500 cases; 30 abstention cases are separated. Category-stratified hash ordering with seed 20260823 assigns 280/92/98 retrieval cases to train/development/test. All profiles share identical cases. Earlier protocol iterations inspected aggregate test results, so the v1.4 test split is held out from v1.4 selection but not pristine across the research process.

The LoCoMo file has 1,986 questions nested in ten conversations: 1,536 category 1–4 questions with explicit evidence and 450 negative-load probes. The v1.4 policy JSON was frozen with SHA-256 before any LoCoMo retrieval run.

## 5.2 Statistics and Cost

We report means plus p50/p95 backend-query latency. LongMemEval comparisons use 5,000 paired bootstrap draws [2]. Questions inside one LoCoMo conversation are dependent, so its 5,000 draws resample whole conversation clusters and then aggregate their paired cases. These intervals represent sampling uncertainty under the public corpora, not systematic label error.

The direct scorer makes no reader-model or LLM-judge calls. This is a central cost advantage over repeated end-to-end trials, but we do not translate it into dollars because hardware and provider pricing are deployment-specific.

**Table 2:** Protocol history. “Rejected” means the candidate was not promoted.

| Version | Outcome      | Change and observation                                                                            |
|---------|--------------|---------------------------------------------------------------------------------------------------|
| v1.0    | Rejected     | Query-only contextual routing lost too much recall.                                               |
| v1.1    | Rejected     | Top-5 scout features did not recover utility confidence.                                          |
| v1.2    | Safe/no gain | Conservative routing collapsed to the static fallback.                                            |
| v1.3    | Global pass  | Fixed 896-token profile passed LongMemEval; contextual routing failed; naive recency was harmful. |
| v1.4    | Mixed        | Relative-budget policy passed development gates, then failed frozen LoCoMo transfer.              |

**Table 3:** Primary results. Recall is macro evidence-session recall. Confidence intervals are paired for LongMemEval and conversation-clustered for LoCoMo.

| Protocol         | Base $R_S$ | Policy $R_S$ | $\Delta R_S$ 95% CI | Base tok. | Policy tok. |
|------------------|------------|--------------|---------------------|-----------|-------------|
| LongMemEval dev. | .8240      | .8500        | $[-.0051, .0607]$   | 1018.8    | 712.2       |
| LoCoMo frozen    | .7403      | .6952        | $[-.0561, -.0335]$  | 168.7     | 122.2       |
| LoCoMo cross-fit | .7403      | .7403        | $[0, 0]$            | 168.7     | 168.7       |

### 5.3 Protocol Evolution Audit

Every change to split, metrics, or profile space increments the protocol and reruns all baselines. Table 2 retains negative iterations.

## 6 Results

### 6.1 Developmental LongMemEval Result

Development selection chooses RELATIVE-75. On 98 held-out cases, macro session recall rises from 0.82398 to 0.85000 and mean injected tokens fall from 1018.77 to 712.17. The recall delta is +0.02602 with paired 95% interval  $[-0.00510, 0.06071]$ ; token delta is  $-306.59$   $[-346.28, -271.77]$ ; and utility is +0.03505  $[0.00431, 0.07048]$ . All promotion checks pass. The result supports method development, not a final cross-domain claim.

### 6.2 Frozen LoCoMo Transfer Failure

The identical policy fails clearly on LoCoMo. Macro session recall changes from 0.74031 to 0.69516, a  $-0.04516$  difference with cluster interval  $[-0.05612, -0.03354]$ . Tokens fall from 168.71 to 122.20 (27.56%), but utility is negative:  $-0.03689$   $[-0.04725, -0.02585]$ . Any- and all-session recall also fall by 4.88 and 4.10 points. The gate correctly rejects the transfer.

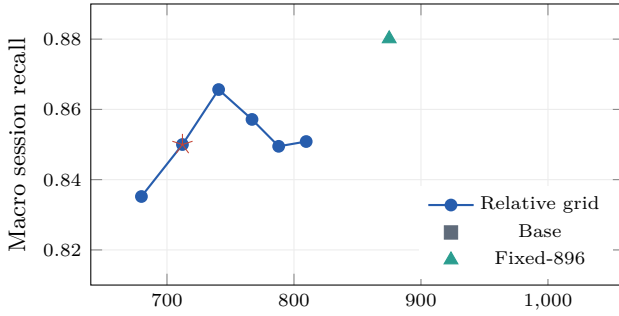
### 6.3 Why Transfer Failed

Figure 2 shows a granularity shift rather than a random miss. A base LongMemEval query contains 1018.8 tokens on average; a base LoCoMo query contains only 168.7. Although the budget is relative, packing is discrete. At 75%, the LoCoMo policy returns 4.21 items instead of the base five, frequently removing an entire evidence turn. On LongMemEval it returns 6.38 shorter items from a deeper pool, improving evidence coverage.

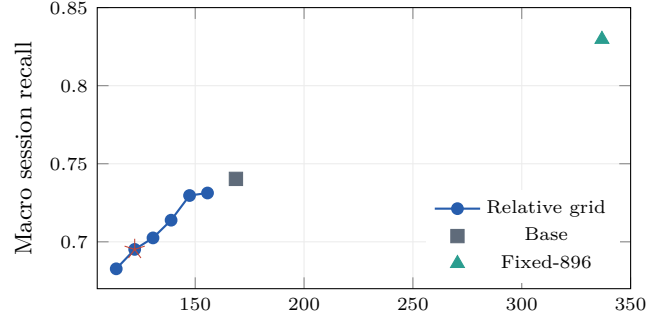
The fixed-896 control further clarifies the trade-off. On LoCoMo it improves macro session recall to 0.82966 but doubles tokens to 336.83. No v1.4 arm simultaneously satisfies the 1-point recall guard and 10% token-saving gate. Changing those thresholds after seeing the result would be protocol shopping, so we retain the failure.

### 6.4 Safe Rejection and Negative Load

In exploratory leave-one-conversation-out calibration, each held-out conversation selects a profile using the other nine. No candidate passes the training-side guards in any fold; all 10/10 choose BASE. The held-out result is therefore



(a) LongMemEval held-out.



(b) LoCoMo external stress test.

**Figure 2:** Macro evidence-session recall versus mean injected tokens. The policy frontier is corpus-dependent. The red star is the LongMemEval-selected RELATIVE-75 profile. It lies above-left of the LongMemEval base, but below-left of the LoCoMo base.

byte-equivalent in policy behavior: no harm and no gain. The system does not count rejection as optimization success.

The method does not solve abstention. On 30 LongMemEval abstention cases, RELATIVE-75 injects 5.93 items versus 5.00 for base. On 450 LoCoMo negative-load probes, it injects 4.19 items and 125.58 tokens versus 5.00 and 173.14 for base. Reducing irrelevant load is useful, but a calibrated zero-injection decision requires explicit sufficiency supervision.

## 6.5 Latency

LongMemEval backend query p50/p95 increases from 0.727/1.867 ms to 1.444/2.620 ms when the candidate limit grows from 5 to 30. LoCoMo increases from 0.382/0.586 ms to 0.845/1.135 ms. These local in-memory measurements are small in absolute terms but roughly double retrieval latency; remote vector stores must be remeasured rather than extrapolated.

## 7 Transfer to Long Programming Tasks

The code separates dataset semantics from the optimizer. An internal adapter would map a task/conversation to a dependency group, the current request to a query, and immutable user messages, tool results, patches, commits, or test failures to provenance units. Explicit “you forgot” feedback is precise but sparse; accepted patches, repeated constraints, reversions, and controlled test outcomes can provide weaker labels with confidence weights.

Three public-data assumptions will fail in programming sessions. First, evidence may be an ordered causal chain rather than an unordered set. Second, tool and repository state may be necessary even when no message contains the answer. Third, later decisions can supersede earlier evidence. A programming protocol should therefore add ordered-chain recall, state snapshot identifiers, supersession labels, and a penalty for injecting stale or contradicted memory. These changes belong in a versioned adapter/metric protocol; they do not require rewriting the policy controller.

The recommended rollout is conservative: collect provenance and direct labels; run base and candidate policies offline; require task-cluster confidence gates; publish a signed/versioned policy; canary the sidepath; and retain immediate base fallback. End-to-end coding-agent trials should follow, not replace, the direct memory gate.

## 8 Limitations and Threats to Validity

**Necessary, not sufficient.** Evidence recall does not show that a reader used evidence correctly. It also cannot certify that an arbitrary truncation preserves the answer-bearing span. We therefore score intact provenance units and do not claim QA accuracy.

**Development contamination.** Protocol v1.4 followed three LongMemEval iterations whose aggregate test results were inspected. Its confidence interval is statistically paired but does not correct researcher adaptivity. We label it developmental and use the frozen LoCoMo result to test transfer.

**Dataset dependence.** LoCoMo has 1,536 evidence questions but only ten independent conversations. Cluster bootstrap respects this structure, yet ten clusters remain a small external sample. LoCoMo annotations and provided image text may contain their own errors.

**Retriever scope.** The public run evaluates L0 SQLite FTS5 only. It does not establish behavior for L1 extraction, embeddings, hybrid RRF, TCVDDB latency, or a real HTTP deployment. The provenance and adapter work enables those experiments but does not substitute for them.

**Cost proxy.** `cl100k_base` is reproducible but may differ from a deployed model’s tokenizer. Latency excludes network and downstream generation. The 250 ms sidepath timeout controls returned behavior but cannot cancel a retriever that does not expose abort.

**Programming claim boundary.** Neither public corpus is a long programming benchmark. We demonstrate a portable method and failure-safe implementation, not production coding-agent improvement.

## 9 Reproducibility, Artifacts, and Ethics

The release records dataset revision/checksum, protocol JSON, split seed, all profile definitions, 5,000 bootstrap draws, structured case outputs, compact result receipts, and the promoted policy checksum. The implementation is on a dedicated branch over MemoryCore v2 source. The structured receipts contain full source revisions; the exact LongMemEval and LoCoMo reruns used `43f3b47` and `abea421`, respectively. Unit tests cover disabled-path equivalence, corrupt policy, baseline mismatch, forced adaptive error, timeout, hard limits, base-relative budget, and L1 provenance round-trip. The complete plugin build is also exercised.

No private user or programming data is used. LongMemEval and LoCoMo are public; LoCoMo’s CC BY-NC 4.0 terms must be respected. Direct evidence logs can expose sensitive user text in a real deployment, so internal reports should store stable source IDs and bounded diagnostics rather than raw content whenever possible.

## 10 Conclusion

EASI-MEM makes memory policy evolution measurable and reversible without rewriting MemoryCore’s retrieval substrate. On LongMemEval it finds a developmentally useful recall/token trade-off. The same checksum-frozen policy fails on LoCoMo, where shorter turns turn a token ratio into an item-deletion policy. A leave-one-conversation-out gate rejects every candidate and returns the exact base path. The honest conclusion is not that one Top- $k$  or token ratio is universally optimal; it is that evidence-aligned feedback, bounded search, versioned receipts, and the ability to reject evolution are all necessary for trustworthy memory self-optimization. The next evaluation should connect the same interfaces to provenance-rich long programming sessions and add state, order, and supersession-aware evidence.

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## A Runtime Configuration Example

```
{
  "recall": {
    "maxResults": 5,
    "adaptive": {
      "enabled": true,
      "policyPath": "policies/easi-mem-v1.4.json",
      "timeoutMs": 250
    }
  }
}
```

The feature defaults to disabled. A relative path is resolved below the plugin data directory. The policy’s declared base must match `maxResults`; a mismatch is logged and falls back.



## B Structured Decision Example

```
{  
  "mode": "adaptive",  
  "policyRevision": "easi-adc30593739a",  
  "selectedProfile": "relative-75",  
  "candidateLimit": 30,  
  "resultLimit": 10,  
  "tokenBudget": 742,  
  "returnedItems": 7,  
  "returnedTokens": 718,  
  "fallback": false,  
  "latencyMs": 2.31  
}
```

Fallback decisions use `mode=base`, `fallback=true`, and a bounded reason indicating policy validation or adaptive-stage failure. No error path silently returns an empty adaptive set as if it were a valid optimization.